综合练习

人员

王承周、许睿谦、王毅博 到课

上周作业检查

上周作业链接: https://vjudge.net/contest/799964

概览 题目 提交状态 排名 (7:05:19:03) 6 条评论 设置 克隆 更新 删除

Rank	Team	Score	Penalty	A 9 / 12	B 9 / 9	C 8 / 12	D 6 / 7	E 5 / 6	F 3 / 5	G 5 / 10	H 3 / 4	I 0 / 0
1	yyycj	8	30615	0:42:13	0:50:38	1:18:09	5:29:08	1:15:43:25	7:00:37:00	5:16:22:38	6:13:12:00	
2	chenxinmiao (陈欣妙)	8	31655	0:41:14	0:48:18	1:31:33 (-1)	1:14:01:00	2:13:43:00	7:00:47:53 (-1)	3:13:05:00 (-2)	7:01:37:02	
3	r111	6	12143	5:31:24	5:54:08	6:21:03 (-1)	1:14:27:01	2:12:49:15		3:13:01:00		
4	wyb1111 (王毅博)	5	9973	5:41:00 (-1)	5:55:12	6:17:37	13:35:00 (-1)			5:13:05:00 (-3)		
5	xuruiqian	5	21674	5:42:09 (-1)	5:54:01	6:25:07	7:03:12:00			7:03:41:00		
6	wcz123bc	4	9909	5:34:23	5:48:19	6:51:03	6:02:56:02				(-1)	
7	chujinxuan (褚锦轩)	4	10345	0:46:41	0:52:01	1:30:43 (-1)		7:00:55:42				
8	lijinshu	3	204	0:45:32	0:53:14	1:25:29 (-1)						
9	junyan007008	3	10321	0:48:31 (-1)	1:18:35						7:01:34:00	
10	7654Fox (隋天乙)	1	10193					7:01:33:10 (-1)				
11	lyc12345bc (刘奕辰)	1	10211						7:01:51:31 (-1)			

本周作业

https://vjudge.net/contest/801929 (课上讲了上周的 E ~ I 题, 课后作业是本周的 A ~ E 题)

课堂表现

今天课上把上周的 5 道题都讲了, 同学们课上没有做完这 5 道题的, 课后要把这些题都要补完。

课堂内容

CF1741E Sending a Sequence Over the Network

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 2e5 + 5;
int w[maxn];
bool f[maxn];

void solve() {
```

```
  int n; cin >> n;
  for (int i = 0; i <= n+2; ++i) f[i] = false;

  for (int i = 1; i <= n; ++i) cin >> w[i];

  f[0] = true;
  for (int i = 1; i <= n; ++i) {
    int l = i - w[i], r = i + w[i];
    if (l-1 >= 0) f[i] |= f[l-1];
    if (r <= n) f[r] |= f[i-1];
  }

//  cout << "------------------------ ";
  cout << (f[n] ? "YES" : "NO") << endl;
}

int main()
{
  int T; cin >> T;
  while (T -- ) solve();
  return 0;
}
```

CF1288D Minimax Problem

二分 + 状压

```
#include <bits/stdc++.h>

using namespace std;

const int N = 3e5 + 5, M = 9;
int w[N][M], f[1<<M];
int n, m;
int resl = 1, resr = 2;

bool check(int mid) {
  memset(f, 0, sizeof(f));
  for (int i = 1; i <= n; ++i) {
    int res = 0;
    for (int j = 0; j < m; ++j) {
      int x = (w[i][j]>=mid);
      res += (x<<j);
    }
    f[res] = i;
  }

  for (int i = 0; i < (1<<m); ++i) {
    for (int j = 0; j < (1<<m); ++j) {
      if (f[i] && f[j] && (i|j)==(1<<m)-1) {
        resl = f[i], resr = f[j];
```

```
        return true;
      }
    }
  }
  return false;
}

int main()
{
  cin >> n >> m;
  for (int i = 1; i <= n; ++i) {
    for (int j = 0; j < m; ++j) cin >> w[i][j];
  }

  int l = 0, r = 1e9+10;
  while (l <= r) {
    int mid = (l + r) / 2;
    if (check(mid)) l = mid+1;
    else r = mid-1;
  }
  cout << resl << " " << resr << endl;
  return 0;
}
```

CF577B Modulo Sum

```
#include <bits/stdc++.h>

using namespace std;

const int N = 1e6 + 5, M = 1e3 + 5;
int w[N];
bool p[M], h[M];

int main()
{
  int n, m; cin >> n >> m;
  for (int i = 1; i <= n; ++i) cin >> w[i];
  if (n >= m) { cout << "YES" << endl; return 0; }

  for (int i = 1; i <= n; ++i) {
    int x = w[i] % m;
    h[x] = true;
    for (int j = 0; j < m; ++j) {
      if (p[j]) h[(x+j)%m] = true;
    }
    memcpy(p, h, sizeof(p));
  }

  cout << (h[0] ? "YES" : "NO") << endl;
```

```
  return 0;
}
```

CF1901D Yet Another Monster Fight

```
#include <bits/stdc++.h>

using namespace std;

const int maxn = 3e5 + 5;
int w[maxn], pre_maxx[maxn], suf_maxx[maxn];

int main()
{
  int n; cin >> n;
  for (int i = 1; i <= n; ++i) {
    cin >> w[i]; pre_maxx[i] = max(pre_maxx[i-1], w[i] + n-i);
  }
  for (int i = n; i >= 1; --i) suf_maxx[i] = max(suf_maxx[i+1], w[i] + i-1);

  int res = 2e9;
  for (int i = 1; i <= n; ++i) res = min(res, max({w[i], pre_maxx[i-1],
suf_maxx[i+1]}));
  cout << res << endl;
  return 0;
}
```

CF1561D2 Up the Strip

dp + 调和级数

针对减法的转移, 跟上个题一致

针对除法的转移, 考虑 f[i]

可以从 f[i*2] ~ f[(i+1)*2-1] 转移而来

可以从 f[i*3] ~ f[(i+1)*3-1] 转移而来

可以从 f[i*4] ~ f[(i+1)*4-1] 转移而来

...

每一段转移可以用前缀和优化, 实现 O(1) 转移

针对 i 来说, 最多有 n/i 段

时间复杂度: n * logn

```cpp
#include <bits/stdc++.h>

using namespace std;

typedef long long LL;
const int maxn = 4e6 + 5;
int mod;
int f[maxn], p[maxn];

int get_sum(int l, int r) { return (p[l]-p[r+1]+mod) % mod; }

int main()
{
  int n; cin >> n >> mod;

  f[n] = 1;
  for (int i = n; i >= 1; --i) {
    f[i] = (f[i] + get_sum(i+1,n)) % mod;

    for (int k = 2; k <= n/i; ++k) {
      int l = i*k, r = min((i+1)*k-1, n);
      f[i] = (f[i] + get_sum(l,r)) % mod;
    }
    p[i] = (p[i+1] + f[i]) % mod;
  }
  cout << f[1] << endl;
  return 0;
}
```